

SCHOOL OF MATERIALS SCIENCE & ENGINEERING

CONTINUOUS ASSESSMENT 2 SEMESTER 2 2021-2022

MS1016 – THERMODYNAMICS OF MATERIAL

14 April 2022

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains THREE (3) short questions and TEN (10) MCQ questions.
2. Answer ALL questions.
3. This is an open book test.
4. All questions have ONLY ONE correct answer for the MCQ questions.

PART I – SHORT QUESTIONS

- (1) Explain the thermodynamic factors affecting the solubility limit of a metallic elements in another metallic element.

$$\Delta G_{mix} = \Delta H_{mix} - T\Delta S_{mix} \quad (10 \text{ marks})$$

Thermodynamic factors affecting the solubility limit of a metallic element in another metallic element is described by the ΔH_{mix} and ΔS_{mix} of the mixture $\rightarrow$ contributing towards ΔG_{mix} .
When bond energy of unlike bonds are stronger than the same atoms / molecules, $\Delta H_{mix} < 0$, and vice versa.
(exothermic)

When $\Delta H_{mix} > 0$, dilute solutions are formed with limited solubility of metallic elements in one another. The solubility limit is affected by thermodynamic variables temperature (T) and pressure (P).
Mostly, pressure is fixed at atmospheric pressure (1 atm) $\rightarrow$ industrial applications.
At higher temperatures, limit of solubility increases, decreasing ΔG_{mix} .
as $-T\Delta S_{mix}$ becomes more dominant at higher temperatures.

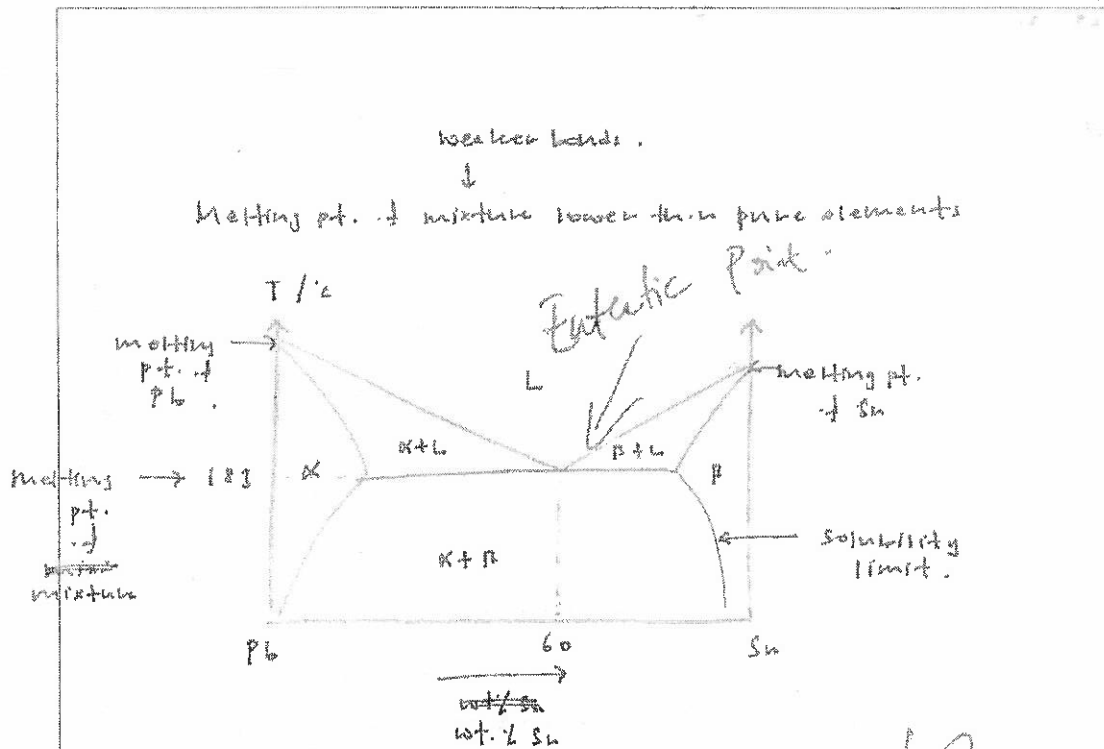
are a driving force for increased entropy (mixing); further reducing Gibbs free energy of mixing: $\Delta G_{mix} = \Delta H_{mix} - T\Delta S_{mix}$.

(62)



2. Solder is an alloy of Pb and Sn. The enthalpy of mixing for the 2 elements, $\Delta H_{mix} > 0$. At 183°C , the alloy with 60% Sn melts directly from solid to liquid phase. Sketch the likely Pb-Sn equilibrium phase diagram and labels the important features.

(20 marks)



$$\Delta G_{mix} = \Delta H_{mix} - T \Delta S_{mix}$$

↑
position

The phase diagram is usually characterized by an eutectic point where solid phase melts directly into liquid phase, in this case at 60% Sn. Such system has higher energy between unlike elements (Pb & Sn) and tend to separate into Pb-rich (α) & Sn-rich (β) phases.

Formation of bonds between unlike atoms requires heat ($\Delta H_{mix} > 0$) resulting in an increase in free energy. endothermic

However, mixing increases entropy and thus reducing (Δ) Gibbs free energy from the equation above.

In order to minimize G , mixing in each component (Pb or Sn) is up to the solubility limit of the components in each other.

3. Describe the use of Ellingham diagram to determine the equilibrium oxidation ^{pressure} ~~temperature~~ of metal. Explain why are metal at lower part of the diagram has a higher affinity for oxygen.

(20 marks)

On the Ellingham diagram, plots of equilibrium pressure of O_2 is superimposed on the plots for ~~the~~ a range of temperatures to determine of the equilibrium of the oxidation reaction of metal to ^{its respective} metal oxides.

To determine the equilibrium oxidation pressure (P_{O_2}) of a metal using the Ellingham diagram, at a specific temperature, a line can be drawn from the "0" dot in the upper left-hand corner of the diagram, to the where the plot for the oxidation of the metal and temperature specified intersects, and then to the P_{O_2} scale of the Ellingham diagram.

At lower part of the diagram, P_{O_2} of the respective metals are lower. This means that these metals, at very low P_{O_2} , can be oxidised to its respective metal oxides (oxidised easily).

$\Rightarrow$ these metals have higher affinity for oxygen than metals with oxidation lines above them (higher P_{O_2})

$\therefore$ these metals will be oxidised first.

E.g.) If P_{O_2} higher than equilibrium P_{O_2} ,

oxidation will proceed: $4C(s) + O_2(g) \rightarrow 2CO_2(g)$

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PART II MCQ QUESTIONS

- (4) Answer all the TEN MCQ questions below. There is only ONE answer for each question.

(50 marks)

$$\Delta H_{mix} < 0$$

- (a) If the enthalpy of mixing of 2 metallic elements is negative, which of the following is most likely to form, at low temperature

- ☒ two dilute solutions
- ☐ an eutectic phase
- ☒ a heterogeneous solution
- ☒ two segregated pure phase
- ☐ an intermetallic phase

ANS: E

- (b) An ordered phase in a binary system will form at the following conditions:

- ☒ low temperature and low bonding energy between same atoms
- ☒ high temperature and low bonding energy between same atoms
- ☐ low temperature and low bonding energy between unlike atoms
- ☒ high temperature and high bonding energy between same atoms
- ☐ low temperature and high bonding energy between unlike atoms

ANS: C

- (c) Which of the following statement is always TRUE?

- ☐ A regular solution has higher enthalpy of mixing than real solution
- ☐ B the enthalpy of mixing for a real solution can be positive or negative
- ☐ C regular solution has lower enthalpy of mixing than real solution
- ☒ D the enthalpy of mixing for a real solution is positive
- ☒ E the enthalpy of mixing for a real solution is negative

ANS: B

- (d) Figure Qd is the free energy curves of a binary system contain A and B. Which of the following statements are correct?

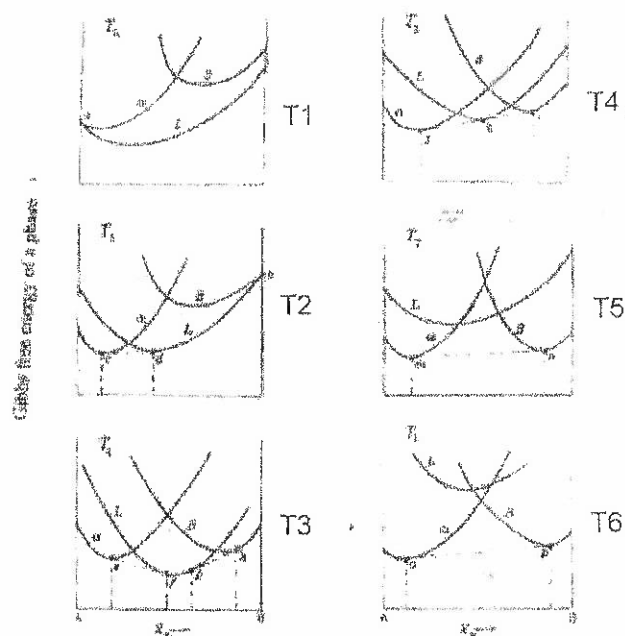


Figure Qd

- ☒ I T_1 is the highest temperature and T_6 is the lowest
- ☒ II T_1 is the lowest temperature and T_6 is the highest
- ☒ III A and B has complete miscibility in each other
- ☒ IV A and B have limited solubility in each other
- ☒ V Melting point of A is $>$ B
- ☒ VI Melting point of A $<$ B
- ☒ VII There is an eutectic point in this binary phase diagram

☒ I, IV, VI, VII

☒ I, III, VI, VII

☒ II, IV, VI, VII

D I, IV, V, VII

☒ II, III, V, VII

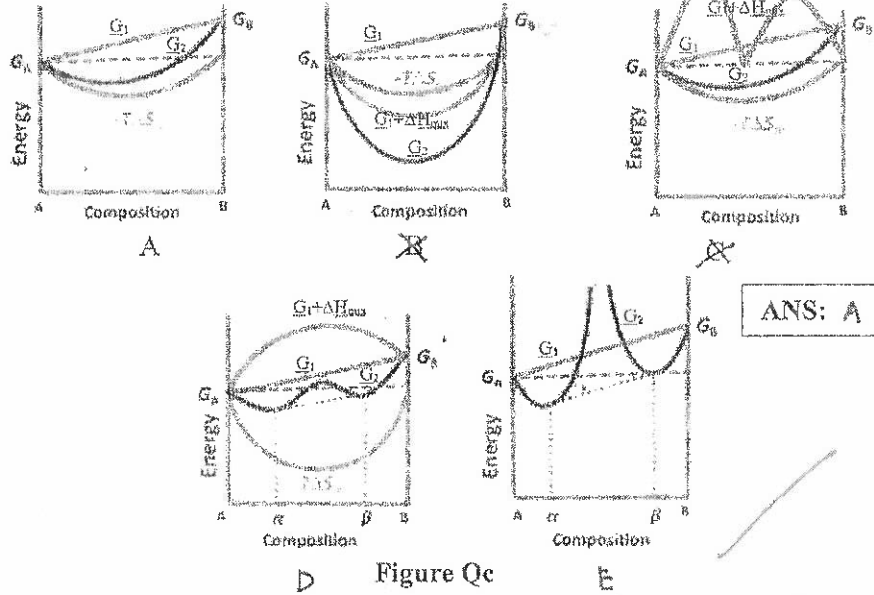
ANS: D

A

X

$$\Delta H_{mix} = 0$$

- (e) Figure Qc are five sets of free energy curves of a binary solution system plotted against the composition. Which set of curves belongs to an ideal solution? G_1 = energy before the components come into contact. G_2 = energy after contact?



- (f) In mixing 2 components together to form an ideal solution at constant temperature and pressure, which of the following is most likely to increase after the components come into contact?

$$\Delta G_{mix} = \Delta H_{mix} - T \Delta S_{mix}$$

- A enthalpy
- B Gibbs free energy
- C Entropy
- D Molar volume
- E temperature

ANS: C

- (g) In the extraction of Si from SiO_2 , from metal oxide, low equilibrium pressure of O_2 can be overcome by the following methods, EXCEPT:

- A introducing a vacuum environment ✓
- B introducing hydrogen ✓
- C increasing oxygen pressure
- D introducing CO ✓
- E introducing Al ✓

ANS: C

(h) Which of the following is TRUE regarding binary solid solution where the components are totally miscible?

- A driving force for ~~clustering~~ occurs in solid solution is entropy
- B decreasing temperature will promote ordering in binary solution
- C ~~increasing~~ temperature will cause ordering in binary solution
- D 3 phases cannot exist in equilibrium
- ~~X~~ Only single phase can exist for all compositions and temperatures

ANS: B

~~(i)~~ Estimate the enthalpy of oxidation reaction of Si.

A -725 kJ

B -750 kJ

C -775 kJ

D -800 kJ

E -825 kJ

$$G = H - TS$$

ANS: E

~~(j)~~ Estimate the equilibrium partial pressure of the oxidation of Mn at 1000°C.

A 10^{-21} atmosphere

B 10^{-22} atmosphere

C 10^{-23} atmosphere

D 10^{-24} atmosphere

E 10^{-25} atmosphere

ANS: D